

PAVEL LIASHKOV

AI/ML Engineer | Agent Systems & LLM Pipelines | Python Expert

pavel.liashkov@protonmail.com | linkedin.com/in/liashkov | github.com/bigbag | kaggle.com/bigbag1983 | liashkov.site

PROFESSIONAL SUMMARY

AI/ML Engineer with 12+ years shipping production systems end-to-end. I build multi-agent LLM pipelines, RAG systems, and evaluation-driven AI applications that serve real customer workflows. Hands-on with prompt engineering, agentic workflows, model evaluation, and ONNX-optimized inference. Kaggle Competitions Expert (top 3%, 3 bronze medals). Active open-source contributor with 2.4M+ PyPI downloads. Hired 20+ engineers through 400+ interviews—comfortable partnering across product, research, and engineering teams.

PROFESSIONAL EXPERIENCE

Backend Engineer — AI/ML & Agent Systems | SplitMetrics | Remote

July 2022 – Present

- Wrote the multi-agent LLM research system end-to-end: PydanticAI-based Planner → 20 Skill Agents → Finalizer, with parallel execution and AG-UI SSE streaming. Powers competitive intelligence across App Store data
- Deep Search pipeline: 4 autonomous agents (Planner, Evaluator, Summarizer, Synthesizer), streak Redis-backed task queue, fact-based convergence detection, user-in-the-loop plan approval with iterative refinement. Reports degrade gracefully on partial failures
- 33-tool FastMCP server + RAG layer: PostgreSQL pgvector for semantic search, DuckDB VSS with ONNX-optimized embeddings. Query latency under 100ms (CPU-optimized INT8 quantization, hybrid HNSW + BM25)
- Added offline LLM evaluation (deterministic checks, subjective judge, hybrid groundedness scoring with difficulty band gating) and a memory extraction service that catches hallucinated app KPIs before they persist
- BERT + ONNX keyword analysis service: sub-second responses at 5K+ req/s
- Wired multi-provider LLM routing across OpenRouter, DeepSeek, Ollama, and OpenAI-compatible endpoints with FallbackModel for automatic model switching
- Ran Snowflake/StarRocks/Airflow data pipelines at 200M+ records/day, 99.9% uptime. AWS infrastructure (ECS, RDS, S3, Bedrock) with Kubernetes and Jenkins CI/CD

Senior Software Engineer | Exness | Limassol, Cyprus

August 2016 – April 2022

- Architected payment processing platform (22+ microservices) with JWS multi-signature validation, state machine invoice lifecycle, and Kafka event-driven reconciliation across 50+ payment providers
- 400+ technical interviews conducted; hired 20+ engineers across backend and platform teams
- Integrated Bitcoin Core RPC and USDT-OMNI for crypto deposits/withdrawals with wallet management and blockchain confirmation tracking
- Implemented PCI DSS-compliant authentication with HashiCorp Vault credential management and JWS cryptographic signing
- Built KYC verification rules engine integrating 12+ MetaTrader servers with SumSub and multi-database connectivity

Software Engineer | Game Stream Joint LLC | Minsk, Belarus

August 2014 – August 2016

- Built microservice-based coupon system and premium shop redesign using Python/Django; developed REST APIs for external platform integration with PostgreSQL and Redis

KEY PROJECTS — AI AGENTS, EVALUATION & DEPLOYMENT

Multi-Agent AI Research System

- Production pipeline with 20 skill agents (6 complex + 14 simple), solo-skill routing bypassing finalizer for single-step tasks, LangSmith tracing, multi-model fallback via OpenRouter—built to debug agent behavior and iterate on prompts in real customer workflows
- Deep Search: async iterative research with convergence detection, plan approval with refinement (up to 3 attempts), and graceful degradation—designed as a reproducible evaluation harness for model quality
- AG-UI protocol + Slack Bolt SDK for conversational interfaces; FastMCP for external tool integration with SSE streaming and run cancellation; Go BFF client

Open Source Libraries — 2.4M+ total downloads, 220K+ monthly on PyPI

- starlette-request-id (1.4M downloads, 120K/mo): ASGI middleware for distributed tracing

- request-id-helper (970K downloads, 100K/mo): Thread-safe request ID tracking for microservices

E-Paper Reader Projects (Open Hardware)

- papyrix-reader (★ 380): ESP32-C3 firmware (~380KB RAM) for e-readers with Knuth-Plass line breaking, Arabic/Thai/CJK multi-script rendering, and streaming font system. C++20, FreeRTOS. epub-to-xtc-converter (★ 90): Browser-based converter with CEngine WASM and Floyd-Steinberg dithering

EDUCATION

Master of Science in Information Technology (MSIT) — 2024

University of the People | GPA: 3.97/4.0

Master's Degree in Theoretical Physics — 2005

Kemerovo State University

CERTIFICATIONS & TRAINING

- Kaggle Competitions Expert — top 3% (rank 1,712 highest), 3 bronze medals in fraud detection, credit risk, and housing market prediction
- Oxford Machine Learning Summer School (OxML 2024): ML theory, Health & Bio, Representation Learning & Generative AI
- Hugging Face AI Agents Course (2025) | Udacity ML Engineer Nanodegree
- CKA: Certified Kubernetes Administrator | Astronomer Certification for Apache Airflow 2 | PSD I (Scrum)
- Yandex Go Developer Professional
- Coursera: Generative AI for Software Development, Agentic AI for Leaders, Data Structures & Algorithms, Self-Driving Cars

TECHNICAL SKILLS

AI/ML & Agents: PydanticAI, OpenAI GPT, Anthropic Claude, PyTorch, BERT, Sentence-Transformers, KeyBERT, ONNX Runtime, FastEmbed, prompt engineering, function calling, agentic workflows, LLM evaluation

Agentic Systems: Multi-agent pipelines, MCP tool servers (FastMCP), AG-UI protocol, OpenRouter, streaq task queue, FallbackModel, workflow automation

RAG & Search: pgvector, DuckDB VSS, hybrid search (HNSW + BM25), vector embeddings, semantic search, BM25 ranking, INT8/AVX512 quantization

Evaluation & Observability: LLM judge scoring, groundedness verification, convergence detection, LangSmith tracing, Pydantic Logfire, Prometheus, Grafana, Sentry

Languages: Python (12+ years), Go (6+ years), SQL, gRPC, Protocol Buffers, REST, GraphQL, MCP, AG-UI

Databases: PostgreSQL (pgvector), Snowflake, StarRocks, Redis, DuckDB, BigQuery, ClickHouse, MySQL

Cloud & Infrastructure: AWS (ECS, RDS, S3, Bedrock), Kubernetes (CKA), Docker, Helm, Jenkins, GitHub Actions, Apache Airflow, Celery, Kafka, RabbitMQ

Security: HashiCorp Vault, PCI DSS, JWS/JWT, RSA/AES hybrid encryption

LANGUAGES

English (Professional working proficiency) | Russian (Native)